

RISHAV SINGH

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SKILLS

Databases :	MySQL, SQLite, MongoDB, FAISS, Chroma
Frameworks / Libraries :	React, PyTorch, TensorFlow/Keras, scikit-learn, LangChain, LangGraph, FastAPI, Flask, OpenCV, pandas, NumPy, NLTK, spaCy, librosa, Streamlit, Gradio
Programming Languages :	Python, SQL, JavaScript, C++, HTML/CSS
AI / GenAI :	LLM API Integration (OpenAI, Gemini), RAG Pipelines, Prompt Engineering, Ollama, Agentic Frameworks, YOLO (v8/v11), Optical Flow, Mel-spectrograms
Soft Skills :	Team Collaboration, Problem Solving and Critical Thinking, Adaptability and Continuous Learning, Attention to Detail, Ownership and Delivery
Tools / Platforms :	Docker, Git, GitHub, Firebase, Azure (exposure), Google Cloud, Power BI

PROJECTS

AI Dashcam Incident Explainer | Individual Project | rishav408 Sep 2025 – Dec 2025

Lead Developer

- Built a local multi-model AI pipeline (YOLO11 + SORT + Farneback optical-flow) to auto-detect incident windows, reducing VLM inference calls by ~80% for real-time edge deployment with zero cloud dependency.
- Analysed keyframes using MiniCPM-V 2.6 for structured JSON fault reports; passed to Mistral 7B for professional insurance-language generation entirely on-device.
- Integrated face blurring (OpenCV DNN), licence-plate anonymisation, PDF export (ReportLab); evaluated on CCD/DoTA/DADA-2000 using BLEU-4 and ROUGE-L.

Technologies / Tools Used : Python, YOLO11, SORT, Optical Flow, MiniCPM-V, Mistral 7B, Ollama, OpenCV, ReportLab

VaniCure — Medical AI Screening Platform | Individual Project | rishav408 Jan 2024 – May 2025

Lead Developer

- Offline-first 3-model ensemble (PANNs CNN14, YAMNet, CNN-BiLSTM) for TB/Asthma screening achieving >0.87 F1 score with no cloud dependency.
- Complete audio preprocessing pipeline (resampling, Mel-spectrograms, MFCCs), FastAPI back-end, and React front-end covering the full AI development lifecycle.
- Multilingual triaging (Hindi/Marathi/Gujarati/English), spatial outbreak prediction, cluster detection, and offline CSV export for rural clinic deployment.

Technologies / Tools Used : Python, PyTorch, TensorFlow/Keras, CNN-BiLSTM, PANNs CNN14, YAMNet, FastAPI, React

NyayaAI — AI Legal Document Assistant | Individual Project | rishav408

Lead Developer

- RAG pipeline (LangChain + Gemini API + FAISS) grounding answers in uploaded documents; stateless Flask REST API with multi-doc support and dynamic per-query context injection.
- Implemented hallucination mitigation via grounded retrieval and dynamic context injection; web-search fallback in a single inference pipeline.

Technologies / Tools Used : Python, RAG, LangChain, Gemini API, FAISS, Flask, NLP

Justify-AI — Multilingual Hate-Speech Detection | Individual Project | rishav408

Developer

- Explainable NLP framework for 5 Indian languages using tokenisation, POS tagging, NER, n-grams, and Naive Bayes classifier with causality extraction and demographic bias auditing.
- Combined risk-score predictor (lexicon + causality + bias signals) served via FastAPI with an interactive analytics dashboard.

Technologies / Tools Used : Python, NLTK, scikit-learn, Naive Bayes, FastAPI, Multilingual NLP

UrbanHive — Housing Society Management System | Team Project | rishav408

Full-Stack Developer

- Power BI KPI dashboard (occupancy, SLA, financials) and ETL pipeline (MongoDB → aggregation → Chart.js) for reporting automation.
- Led 4-member Agile team translating stakeholder requirements into analytics views through iterative UAT-style feedback cycles.

Technologies / Tools Used : React, Node.js, MongoDB, SQL, Power BI, Chart.js

EDUCATIONS

MIT WORLD PEACE UNIVERSITY | Pune Jul 2021 – Present

B.Tech in Computer Engineering and Technology (Artificial Intelligence and Data Science) CGPA : 7.43/10

AWARDS & ACHIEVEMENTS

Anthropic AI Certifications (AI Fluency for Students; Claude Code in Action) 2026

Anthropic

- Completed two official Anthropic certifications covering responsible AI, LLM fundamentals, agentic workflows, tool use, and production-grade LLM application development.

Artificial Intelligence Fundamentals 2025

IBM SkillsBuild

- Completed IBM's AI Fundamentals program covering ML/DL theory, model building, evaluation, and practical AI application deployment.

Introduction to LangChain (Python) | Critical Thinking & Problem Solving 2024

LangChain Academy | Rochester Institute of Technology (RITx)

- LangChain Academy: built LLM chains, agents, and RAG pipelines. RITx: structured reasoning, analytical decision-making, and engineering problem-solving frameworks.

PROFESSIONAL SUMMARY

Final-year B.Tech (AI & Data Science) student at MIT World Peace University with hands-on experience building LLM-powered applications, RAG pipelines, multi-model inference systems, and REST APIs using Python. Developed production-ready AI projects including a local dashcam incident analyser (YOLO11 + on-device LLMs, zero cloud), a medical screening platform (3-model ensemble, FastAPI, React), and an AI legal assistant (RAG + LangChain + Gemini). Familiar with PyTorch, TensorFlow, Docker, Azure, and agentic AI frameworks. Actively seeking an AI/ML internship to contribute to real-world intelligent systems and grow within a collaborative engineering team.